M5 · Complement made easy

Innate immunity lecture pack · EAACI/UEMS Knowledge Exam preparation

After this module you can

- Explain in one sentence what complement does and why it amplifies so fast.
- Name the trigger, sensor and proteases of the classical, lectin and alternative pathways.
- Write the C3 convertase and C5 convertase of each pathway.
- Describe the assembly of the membrane attack complex (MAC), step by step.
- Link each complement fragment to its function and receptor (CR1–CR4, C5aR1).
- Place each regulator at the exact step where it acts, and link it to the disease in M6.

MAC	membrane attack complex (C5b-9)
C1-INH	C1 inhibitor
MBL	mannose-binding lectin
MASP	MBL-associated serine protease
CRP	C-reactive protein
CR1–CR4	complement receptor type 1 to type 4
C5aR1	C5a receptor 1
C4BP	C4-binding protein
MCP	membrane cofactor protein (CD46)
DAF	decay-accelerating factor (CD55)
GPI	glycophosphatidylinositol (membrane anchor)
ICAM-1	intercellular adhesion molecule 1
EBV	Epstein-Barr virus
HAE	hereditary angioedema
PNH	paroxysmal nocturnal hemoglobinuria
LAD	leukocyte adhesion deficiency

1. What complement is — in plain words

Complement is a set of plasma proteins. It is a major effector arm of humoral (antibody) immunity and also an important effector arm of innate immunity. Abbas p.300

Most complement proteases circulate as inactive zymogens. A zymogen becomes active only when another protease cleaves it. Janeway p.49 Each active protease then cleaves many molecules of the next zymogen. Thus the signal grows at every step. Janeway p.49

> **Analogy**
>
> Think of a row of dominoes where each domino knocks over ten more. One small trigger on a bacterium quickly becomes thousands of active molecules on that same bacterium.

Complement does four main jobs. You will meet each one again in section 7:

1. **Opsonization** — C3b coats the microbe so that phagocytes eat it.
2. **Inflammation** — small fragments (C3a, C5a) call in cells and activate mast cells.
3. **Lysis** — the membrane attack complex (MAC) punches holes in some microbes.
4. **Help for adaptive immunity** — clearance of immune complexes and B-cell activation.

Abbas p.313; Abbas p.307; Abbas p.314

Name rules that make complement easy

- When a protein is cleaved, the **larger** fragment gets the letter **b** and the smaller fragment gets **a** (C3 → C3a + C3b). C2 is the only exception. Janeway p.49
- For C2, the larger fragment that stays bound is **C2a**. The small released fragment is **C2b**. Both books use this rule, so the classical C3 convertase is written **C4b2a**. Abbas p.305
- The numbers follow the order of discovery, not the order of action. The classical sequence is C1 → C4 → C2 → C3 → C5 → C6 → C7 → C8 → C9. Janeway p.49

> **Exam pearl**
>
> "Small a, big b — except C2." If a question writes C4b2a, the "2a" is the large, enzymatic piece of C2.

2. The big picture: three ways in, one common road

Three pathways start complement. Abbas p.301

- The **classical pathway** starts on antibody bound to antigen.
- The **lectin pathway** starts when lectins bind sugars on microbes.
- The **alternative pathway** starts on microbial surfaces without antibody.

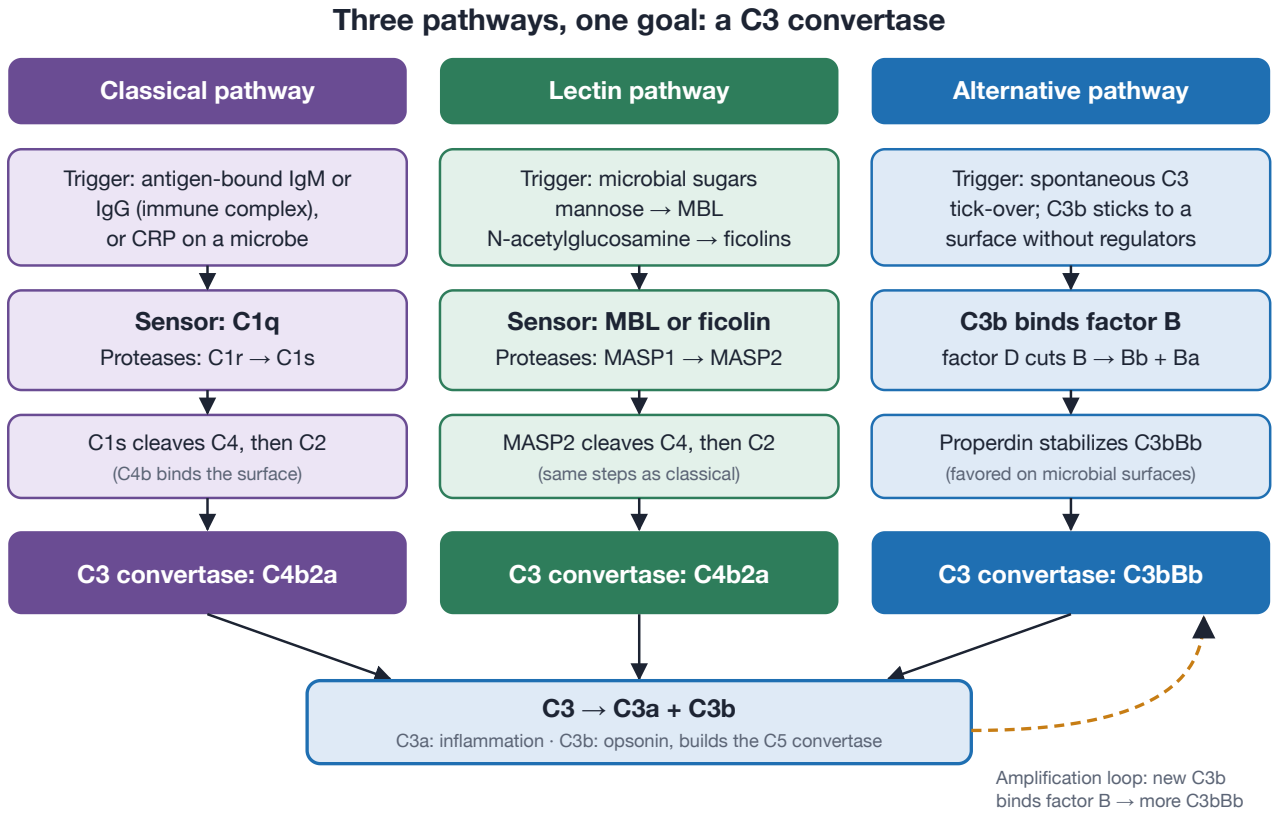
The alternative and lectin pathways are innate effector mechanisms. The classical pathway is a major mechanism of adaptive humoral immunity. Abbas p.301 However, Janeway notes that all three pathways can start without antibody, as part of innate immunity. Janeway p.71

All three pathways make the same key enzyme activity: a **C3 convertase**. The C3 convertase cleaves C3 into C3a and C3b. C3b then binds covalently to the target surface. This is the central event of complement activation. Abbas p.301

Cleavage exposes a reactive thioester bond in C3b. If C3b does not bond to a nearby surface, water hydrolyzes the thioester and C3b becomes inactive. Janeway p.52 This keeps complement on the surface where it started.

Normal host cells carry regulatory proteins that stop complement. Microbes lack these proteins. Therefore complement proceeds on microbes but not on our own cells. Abbas p.301

> **Analogy**
>
> C3b is like wet paint with a "please eat me" label. The paint sticks only if it lands on a surface quickly; otherwise it dries and is useless. Our own cells carry "paint remover" (the regulators). Bacteria do not.



D15. The three pathways differ only in how they start. Classical and lectin pathways build the same C3 convertase (C4b2a); the alternative pathway builds C3bBb. Every C3b made by any pathway can enter the alternative-pathway amplification loop (dashed). Abbas p.301; Abbas p.305; Abbas p.306; Abbas p.307; Janeway p.54; Abbas p.302

3. The classical pathway — step by step

The classical pathway starts when C1 binds the CH2 domain of IgG or the CH3 domain of IgM. The antibody must already be bound to antigen. Abbas p.305

C1 has three parts. C1q is the sensor that binds antibody. C1r and C1s are serine proteases. Abbas p.305

1. C1q binds at least two Ig heavy chains. Free, circulating antibody cannot do this, so only antigen-bound antibody triggers the pathway. Abbas p.305
2. Bound C1q activates C1r. C1r cleaves and activates C1s. Abbas p.305
3. C1s cleaves C4. C4b binds covalently to the surface. Abbas p.305
4. C2 binds the surface C4b. C1s cleaves C2. The large C2a fragment stays on C4b. Abbas p.305
5. The C4b2a complex is the classical pathway **C3 convertase**. Abbas p.305

6. C4b2a cleaves C3. Some new C3b binds to the convertase itself. C4b2a3b forms, the classical **C5 convertase**. Abbas p.306

Which antibodies and molecules trigger C1?

- **IgM**: one antigen-bound IgM can bind two or more C1q molecules. Therefore IgM fixes complement better than IgG. Abbas p.305
- **IgG**: IgG1 and IgG3 are the most efficient human subclasses. Abbas p.305 IgG2 is weak, and IgG4 does not activate complement. Abbas p.305
- **Pentraxins**: C1q also binds C-reactive protein (CRP) and serum amyloid P. C1q also binds apoptotic bodies. Abbas p.305

> **Exam pearl**
>
> A single IgG molecule cannot activate C1, because C1q needs two Fc regions close together. One antigen-bound IgM is enough. IgG4 does not fix complement. Abbas p.305

4. The lectin pathway — the classical pathway's twin

The lectin pathway starts when plasma mannose-binding lectin (MBL) or ficolins bind polysaccharides on microbes. Abbas p.306 MBL binds mannose residues. Abbas p.306 Ficolins (L-, H- and M-ficolin) bind N-acetylglucosamine-containing glycans with a fibrinogen-like domain. Abbas p.306

1. MBL or a ficolin binds sugars on the microbe. Abbas p.306
2. MASP1 may activate MASP2. MASP2 cleaves C4 and C2, just like C1s. Abbas p.307
3. The result is the same C3 convertase as in the classical pathway: C4b2a. Janeway p.54

> **Analogy**
>
> The classical and lectin pathways are two different keys for the same lock. C1q reads "antibody"; MBL and ficolins read "microbial sugar". After the key turns, the machinery (C4, C2, C4b2a) is identical.

5. The alternative pathway — always idling, ready to amplify

Plasma C3 breaks down spontaneously at a low rate, about 1–2% of total C3 per hour. This is called C3 **tick-over**. Abbas p.301

1. Tick-over gives C3(H_2O). C3(H_2O) binds factor B. Janeway p.58
2. Factor D cleaves the bound factor B. A short-lived fluid-phase C3 convertase, C3(H_2O)Bb, forms. Janeway p.58
3. This enzyme makes C3b. Some C3b lands on a nearby surface. Janeway p.58

4. Surface C3b binds factor B. Factor D cleaves it; Ba is released and Bb stays. Abbas p.301
5. C3bBb is the alternative pathway **C3 convertase**. Abbas p.301
6. Properdin binds and stabilizes C3bBb. Properdin attaches preferentially to microbial surfaces. Abbas p.302
7. A second C3b joins the convertase. The complex with one Bb and two C3b (C3bBb3b) is the alternative **C5 convertase**. Abbas p.303

Factor D is special. It is the only activating protease of complement that circulates as an active enzyme, not as a zymogen. Janeway p.61

The amplification loop

Each C3bBb makes more C3b, and each new C3b can form more C3bBb. Thus the alternative pathway C3 convertase amplifies complement activation started by *any* of the three pathways. Abbas p.301

> **Analogy**
>
> Tick-over is a pilot light that always burns a little. On our own cells, regulators blow the flame out at once. On a bacterium, nothing stops it, so the pilot light becomes a fire that feeds itself (the amplification loop). Properdin is the wind shield that keeps the flame alive.

> **Exam pearl**
>
> Homologous pairs: C3 ↔ C4 and factor B ↔ C2. Abbas p.306 So C3bBb is the "alternative twin" of C4b2a: C3b plays the role of C4b, and Bb plays the role of C2a.

6. Convertases and the membrane attack complex (MAC)

The four convertases at a glance

Pathway	C3 convertase	C5 convertase	Source
Classical	C4b2a	C4b2a3b	Abbas p.305; Abbas p.306
Lectin	C4b2a (same as classical)	C4b2a3b	Janeway p.54; Janeway p.62
Alternative	C3bBb (fluid phase: $C3(H_2O)Bb$)	C3bBb3b (Janeway: $C3b_2Bb$)	Abbas p.301; Janeway p.58; Abbas p.303; Janeway p.62

A C5 convertase is simply a C3 convertase plus one extra C3b. C5 binds to this C3b. Then C2a or Bb cleaves C5 into C5a and C5b. Janeway p.62

Exam pearl — nomenclature

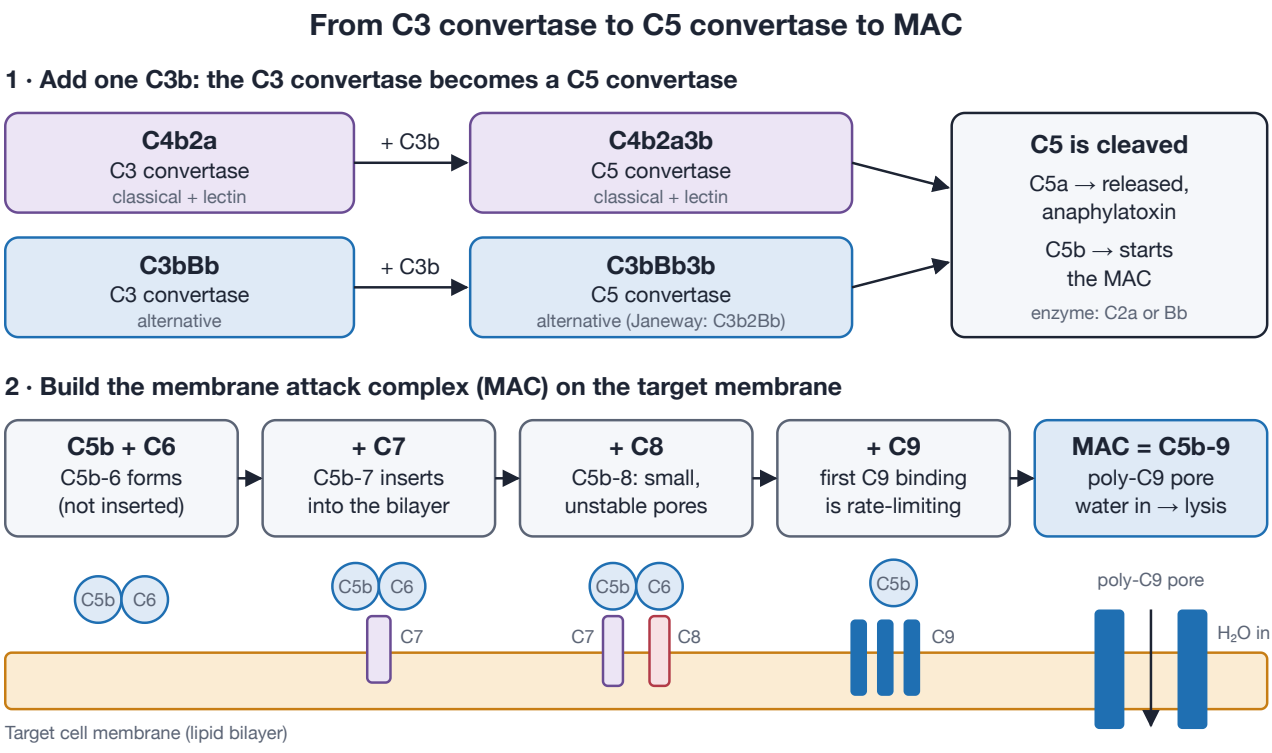
Abbas and Janeway both write the classical/lectin C3 convertase as C4b2a and the C5 convertase as C4b2a3b. Abbas p.305; Janeway p.62 For the alternative C5 convertase, Abbas describes one Bb with two C3b, and Janeway writes $C3b_2Bb$. These names describe the same complex.

Abbas p.303; Janeway p.62

Building the MAC

C5b starts the late (terminal) steps. Abbas p.307 Unlike C3b and C4b, C5b does not bind covalently to the cell surface. Janeway p.62

1. C5b binds C6, and then C7. The C5b-7 complex inserts into the lipid bilayer. Abbas p.307
2. C8 joins. C5b-8 makes small, unstable pores (0.4–3 nm). Abbas p.307
3. The first C9 binds. This is the rate-limiting step. Abbas p.307
4. More C9 molecules polymerize (poly-C9). The complete pore is C5b-9, the **membrane attack complex (MAC)**. Abbas p.307
5. Water enters through the pore. The cell swells and ruptures (osmotic lysis). Abbas p.307



- Abbas: up to 18 C9 per pore. Janeway: 10–16 C9. Same idea, different counts.
- C5b-7 and C5b-8 already enter the membrane; C9 completes the channel.
- Water and ions flow in, the cell swells and bursts (osmotic lysis).

D16. Top: one extra C3b turns each C3 convertase into a C5 convertase. Bottom: C5b collects C6, C7, C8 and many C9 molecules to build the MAC pore. Abbas p.306; Abbas p.303; Janeway p.62; Abbas p.307; Janeway p.66

> **Source difference**
>
> The number of C9 molecules per MAC differs: Abbas gives "up to 18", Janeway gives 10–16. Abbas p.307; Janeway p.66 An exam will not ask for the exact number. Remember "C5b-9 with poly-C9".

How important is lysis?

MAC lysis is critical against only a few pathogens. The classic example is *Neisseria*, which has a thin cell wall. Abbas p.314 Deficiency of C5–C9 is associated with susceptibility to *Neisseria* only. Janeway p.66 → Details in M6.

7. What complement does — the functions

7.1 Opsonization (the most important job)

The most important action of complement is to help phagocytes take up and destroy pathogens. Janeway p.62 C3b and C4b bind CR1. iC3b, a cleaved form of C3b, binds CR3 (MAC-1) and CR4. Abbas p.313 By itself, CR1 is inefficient at triggering phagocytosis of C3b-coated microbes. Simultaneous IgG binding to Fcγ receptors enhances it. Abbas p.313 In contrast, iC3b binding to CR3 is sufficient on its own to trigger phagocytosis. Janeway p.64

> **Clinical correlation**
>
> IgM against capsular polysaccharides activates the classical pathway. Phagocytes then clear pneumococci and meningococci in the spleen. Abbas p.313 Therefore, people without a spleen are susceptible to disseminated pneumococcal and meningococcal septicemia. Abbas p.313

7.2 Inflammation: the anaphylatoxins C3a, C4a, C5a

C5a, C4a and C3a bind to mast cells. They cause degranulation with release of histamine and other vasoactive mediators. Abbas p.313 These peptides are called **anaphylatoxins**, because the mast cell reactions they trigger resemble anaphylaxis. Abbas p.313

- **Potency:** C5a is the most potent mediator of mast cell degranulation. C3a is about 20-fold weaker, and C4a about 2500-fold weaker. Abbas p.314
- **Neutrophils:** C5a stimulates motility and firm adhesion to endothelium. At high doses, it stimulates the respiratory burst. Abbas p.313
- **Eosinophils:** C5a is a chemoattractant for eosinophils. Abbas p.473
- **Receptor:** C5aR1 is a G protein-coupled receptor. It is on neutrophils, eosinophils, basophils, monocytes, macrophages, mast cells and other cells. Abbas p.309
- **Off switch:** plasma carboxypeptidases remove the C-terminal arginine. The des-Arg forms keep only about 10% of the activity. Abbas p.312

> **Clinical correlation — allergy**
>
> C3a and C5a can degranulate mast cells *without* IgE. Abbas p.470 In mouse studies of bacterial infection, alternative pathway activation releases C5a, which directly triggers mast cell degranulation. Abbas p.482 Large systemic amounts of anaphylatoxins cause circulatory collapse, a shock-like syndrome that resembles a systemic IgE-mediated allergic reaction. Janeway p.64

> **Analogy**
>
> C3b is the "eat me" label that stays on the bacterium. C3a and C5a are the smoke that drifts away and sets off the fire alarm (mast cells, neutrophils, eosinophils).

7.3 Lysis

The MAC forms a pore, and water enters the cell. Abbas p.307 See section 6 and diagram D16.

7.4 Immune-complex clearance

CR1 on erythrocytes binds circulating immune complexes that carry C3b and C4b. Erythrocytes carry these complexes to the liver and spleen. There, phagocytes remove them. Abbas p.307

7.5 Help for B cells

Factor I cleaves C3b to iC3b, C3d and C3dg. These fragments do not continue the cascade, but they bind receptors on phagocytes and B cells. Abbas p.312 C3d on an antigen binds CR2 on B cells and facilitates B-cell activation. Abbas p.314 On B cells, CR2 (CD21) forms a co-receptor complex with CD19 and CD81. Abbas p.308

> **Analogy**
>
> C3d is a "second signature" on the antigen. A B cell that sees both the antigen (with its B-cell receptor) and the C3d tag (with CR2) gets a much louder "go" signal.

Complement receptors — summary table

Receptor	Main ligand(s)	Main cells / role	Source
CR1 (CD35)	C3b, C4b	Phagocytes; erythrocytes carry immune complexes; also a regulator (cofactor and decay)	Abbas p.313; Abbas p.307; Janeway p.60
CR2 (CD21)	C3d, C3dg, iC3b	B cells (with CD19, CD81); EBV receptor	Abbas p.308; Abbas p.310
CR3 (MAC-1, CD11b/CD18)	iC3b; also ICAM-1	Integrin; phagocytosis and adhesion to endothelium	Abbas p.309; Janeway p.64
CR4 (CD11c/CD18)	iC3b	Integrin; CD11c marks dendritic cells	Abbas p.309
C5aR1	C5a	G protein-coupled receptor on granulocytes, mast cells, monocytes and others	Abbas p.309

> **Source difference**
>
> Abbas lists iC3b as a weak CR1 ligand (C3b > C4b > iC3b). Abbas p.310 Janeway says iC3b binds CR2, CR3 and CR4, but not CR1. Janeway p.64 For the exam, link iC3b mainly to CR3 and CR4.

CR3 and CR4 share the CD18 chain. Mutations in CD18 remove both receptors and cause leukocyte adhesion deficiency (LAD). Abbas p.315 → See M4.

8. Regulators — the brakes, and where each one acts

The reactive thioester of C3b and C4b cannot tell a host cell from a microbe. Janeway p.68 Therefore host cells need brakes. Regulators act at three levels: they block C3 convertase formation, break down C3 and C5 convertases, and inhibit the MAC. Abbas p.311 Microbes lack the membrane regulators. Abbas p.301

> **Analogy**
>
> Think of a car with several brakes. C1-INH is the hand brake at the start. DAF, factor H and C4BP pull the convertase apart (they "accelerate decay"). Factor I is the scissors, but it cuts only when a helper (cofactor) holds C3b or C4b. CD59 is the last brake, at the MAC.

8.1 At the start: C1 inhibitor (C1-INH)

- C1-INH is a serpin. Active C1r and C1s cleave it, it binds them covalently, and C1r2s2 dissociates from C1q. This stops the classical pathway. Abbas p.311
- C1-INH also inactivates MASP2, so it dampens the lectin pathway. Abbas p.311
- C1-INH also inhibits kallikrein and coagulation factor XII. Both enzymes promote bradykinin formation. Abbas p.311

→ **M6 link:** C1-INH deficiency causes hereditary angioedema (HAE), an autosomal dominant disease. Abbas p.311 The edema mediators include C2 kinin and bradykinin. Abbas p.311

8.2 At the convertases: plasma regulators

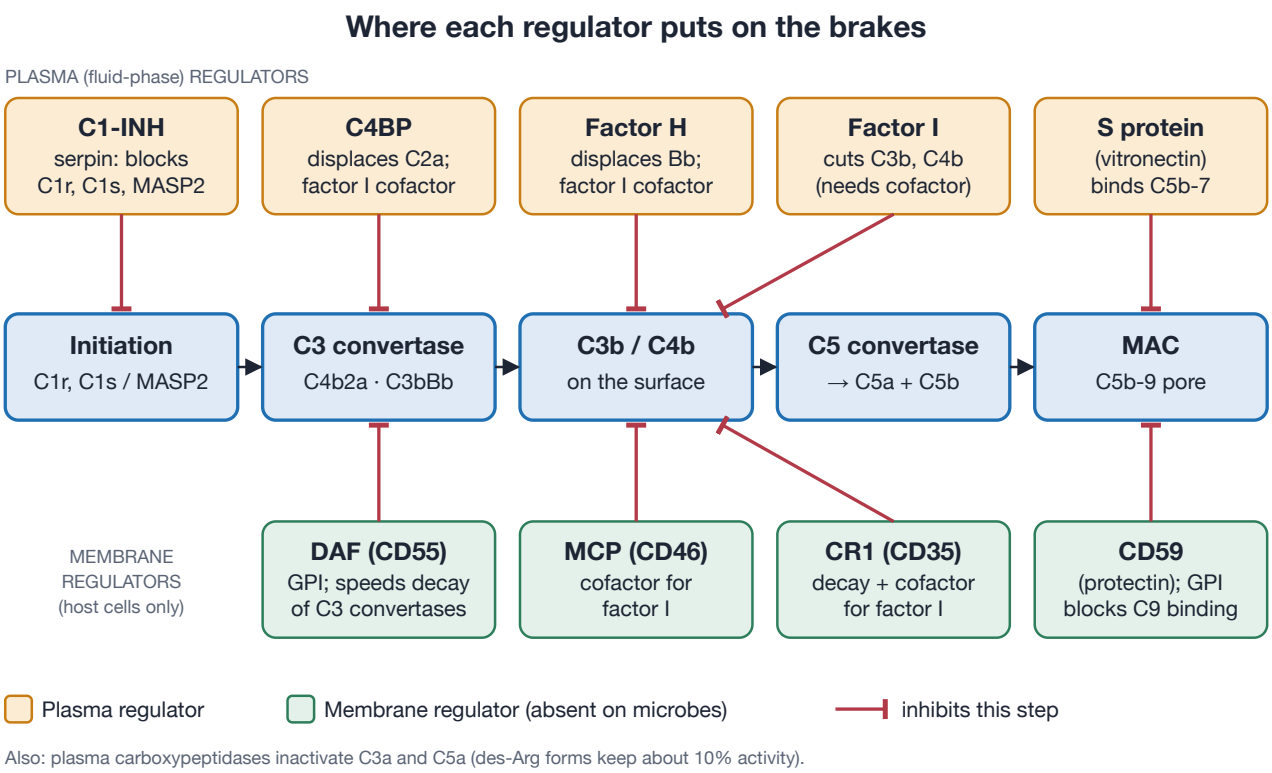
- **Factor H** binds C3b, displaces Bb, and is a cofactor for factor I. Abbas p.310 It blocks only Bb, so it regulates the alternative pathway but not the classical pathway. Abbas p.311 Sialic acid-rich host surfaces favor factor H binding over factor B. Abbas p.311 → **M6 link:** factor H is the gene most often mutated in complement-mediated hemolytic-uremic syndrome. Abbas p.315
- **C4-binding protein (C4BP)** binds C4b, displaces C2, and is a cofactor for factor I cleavage of C4b. Abbas p.310 → **M6 link:** C4BP deficiency: Not covered in provided sources — needs expert input.
- **Factor I** is a plasma serine protease. It cleaves C3b and C4b, using factor H, MCP, C4BP or CR1 as cofactor. Abbas p.310 It circulates in active form, but cuts only C3b or C4b that is bound to a cofactor. Janeway p.68 → **M6 link:** factor I deficiency depletes plasma C3 and causes pyogenic infections. Abbas p.315

8.3 At the convertases: membrane regulators

- **Decay-accelerating factor (DAF, CD55)** displaces C2a from C4b and Bb from C3b, so it dissociates both C3 convertases. Abbas p.310 → **M6 link:** DAF is GPI-linked; loss of GPI anchors removes DAF and CD59 and causes paroxysmal nocturnal hemoglobinuria (PNH). Abbas p.311
- **Membrane cofactor protein (MCP, CD46)** is on leukocytes, epithelial cells and endothelial cells. It is a cofactor for factor I cleavage of C3b and C4b. Abbas p.310 → **M6 link:** MCP mutations also occur in complement-mediated hemolytic-uremic syndrome. Abbas p.315
- **CR1 (CD35)** both displaces Bb (decay) and acts as a cofactor for factor I. Janeway p.60

8.4 At the MAC

- **CD59 (protectin)** is a GPI-linked membrane protein. It enters the assembling MAC after C5b-8 insertion and blocks the addition of C9. Abbas p.312; Janeway p.70 → **M6 link:** CD59 loss contributes to hemolysis in PNH. Abbas p.311
- **S protein (vitronectin)** in plasma binds soluble C5b-7. It stops C5b-7 from inserting into bystander cell membranes. Abbas p.312; Janeway p.70



D17. Plasma regulators (top) and host-cell membrane regulators (bottom) act at three levels: initiation, the C3b/C4b-containing convertases, and the MAC. Microbes lack the membrane regulators, so complement proceeds on them.

Abbas p.301; Abbas p.311; Abbas p.310; Janeway p.68; Abbas p.312; Janeway p.70; Janeway p.60

Regulators — one table to learn

Regulator	Where	Acts on	What it does	M6 link
C1-INH	Plasma	C1r, C1s, MASP2	Serpin; removes C1r2s2 from C1q Abbas p.311	HAE Abbas p.311
C4BP	Plasma	C4b	Displaces C2; factor I cofactor Abbas p.310	Not covered in provided sources — needs expert input.
Factor H	Plasma	C3b	Displaces Bb; factor I cofactor; alternative pathway only Abbas p.310; Abbas p.311	Hemolytic-uremic syndrome Abbas p.315
Factor I	Plasma	C3b, C4b	Cleaves them; needs a cofactor Abbas p.310; Janeway p.68	C3 depletion, pyogenic infections Abbas p.315
MCP (CD46)	Membrane	C3b, C4b	Factor I cofactor Abbas p.310	Hemolytic-uremic syndrome Abbas p.315
DAF (CD55)	Membrane (GPI)	C4b2a, C3bBb	Dissociates C3 convertases Abbas p.310	PNH Abbas p.311
CR1 (CD35)	Membrane	C3b, C4b	Decay and factor I cofactor Janeway p.60	—
CD59 (protectin)	Membrane (GPI)	C5b-8	Blocks C9 binding; stops the MAC Abbas p.312; Janeway p.70	PNH Abbas p.311
S protein (vitronectin)	Plasma	Soluble C5b-7	Prevents bystander insertion Abbas p.312; Janeway p.70	—

Exam pearl — decay vs cofactor

Decay accelerators pull the convertase apart: DAF, CR1, factor H (displace Bb) and C4BP (displaces C2). **Cofactors** help factor I cut C3b or C4b: MCP, CR1, factor H, C4BP. CR1 and factor H do both. Janeway p.60; Abbas p.310

9. How microbes escape complement (brief)

- Thick cell walls of Gram-positive bacteria and some fungi prevent binding of complement proteins such as the MAC. Abbas p.316
- Many pathogens recruit host regulators. For example, surface sialic acid recruits factor H. Abbas p.316

5 key points

1. Three pathways (classical, lectin, alternative) all lead to a C3 convertase; cleavage of C3 is the central event. Abbas p.301

2. Classical and lectin pathways share the C3 convertase C4b2a; the alternative C3 convertase is C3bBb. Adding one C3b gives the C5 convertases C4b2a3b and C3bBb3b.
   Abbas p.305; Janeway p.54; Abbas p.301; Abbas p.306; Abbas p.303
3. The alternative pathway amplifies all three pathways; properdin stabilizes C3bBb, and factor D circulates already active. Abbas p.301; Abbas p.302; Janeway p.61
4. C3b and iC3b opsonize (CR1, CR3, CR4); C5a is the strongest anaphylatoxin and degranulates mast cells without IgE; the MAC (C5b-9) lyses thin-walled *Neisseria*.
   Abbas p.313; Abbas p.314; Abbas p.470
5. Regulators act at three levels: C1-INH at C1/MASP2; DAF, MCP, CR1, factor H, C4BP and factor I at the convertases; CD59 at the MAC. Abbas p.311; Janeway p.60; Abbas p.310; Abbas p.312

3 exam traps

1. Factor H regulates the **alternative** pathway only; it does not displace C2a from C4b. The classical-pathway plasma regulator of C4b is C4BP. Abbas p.311; Abbas p.310
2. CD59 does not block convertases. It blocks C9 binding at C5b-8. DAF is the GPI-linked regulator that dissociates the C3 convertases. Abbas p.312; Abbas p.310
3. Free circulating IgG or IgM does not activate C1. C1q needs two Fc regions on antigen-bound antibody; IgG4 does not fix complement. Abbas p.305

Practice: M5 questions in the quiz (IDs M5-A1 … M5-K2).

Sources: Abbas, Cellular and Molecular Immunology, 11th ed., pp. 300–316 (complement), 470, 473, 482 (mast cells and anaphylatoxins); Janeway's Immunobiology, 8th ed., pp. 49–71 (complement and innate immunity). · Fact IDs refer to `ledger/facts.csv`.